TECHAVO'S

LORAWAN SENSOR NODE

TSDTU118

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OVERVIEW

LSN50v2 is a Long Range LoRaWAN Sensor Node. It is designed for outdoor use and powered by a Li/SOCl2 battery for long-term use, power consumption, and secure data transmission. It is designed to help developers rapidly deploy industrial-level LoRaWAN and IoT solutions. LSN50 v2 helps users turn the idea into a practical application and make the Internet of Things a reality. It is easy to program, create, and connect your things everywhere. LSN50v2 is based on SX1276/SX1278 allows the user to send data and reach extremely long ranges at low data-rates. It provides ultra-long-range spread spectrum communication and high interference immunity while minimizing current consumption. It targets professional wireless sensor network applications such as irrigation systems, smart metering, smart cities, smartphone detection, building automation, etc.

OVERVIEW

LSN50v2 uses an STM32l0x chip from ST; STML0x is the ultra-low-power STM32L072xx microcontrollers that incorporate the connectivity power of the universal serial bus (USB 2.0 crystal-less) with the high-performance ARM® Cortex®-M0+ 32-bit RISC core operating at a 32 MHz frequency, a memory protection unit (MPU), high-speed embedded memories (192 Kbytes of Flash program memory, 6 Kbytes of data EEPROM and 20 Kbytes of RAM) plus an extensive range of enhanced I/Os and peripherals.

LSN50v2 is an open source product, it is based on the STM32Cube HAL drivers and lots of libraries can be found in ST site for rapid development.

FEATURES

- STM32L072CZT6 MCU
- Ultra-low power consumption
- MDK-ARM Version 5.24a IDE
- LoRa™ Modem
- Preamble detection
- Baud rate configurable
- CN470/EU433/KR920/US915
- Open-source hardware/software
- Available Band:433/868/915/920 Mhz
- IP68 Waterproof Enclosure
- EU868/AS923/AU915
- Pre-load bootloader on USART1/USART2
- SX1276/78 Wireless Chip
- 2x12bit ADC, 1x12bit DAC
- I2C,LPUSART1,USB
- 18xDigital I/Os
- AT Commands to change parameters
- 4000mAh Battery for long-term use

SPECIFICATIONS

MCU Side

- MCU: STM32L072CZT6
- Flash:192KB
- Clock Speed: 32Mhz
- RAM:20KB
- EEPROM:6KB

Absolute Maximum Ratings:

- I/O pins: 0.5v ~ VCC+0.5V

Common DC Characteristics:

- Supply Voltage: 2.1v ~ 3.6v
- Operating Tempature: -40 ~ 85°C
- I/O pins: Refer to STM32L072 datasheet

Power Consumption:

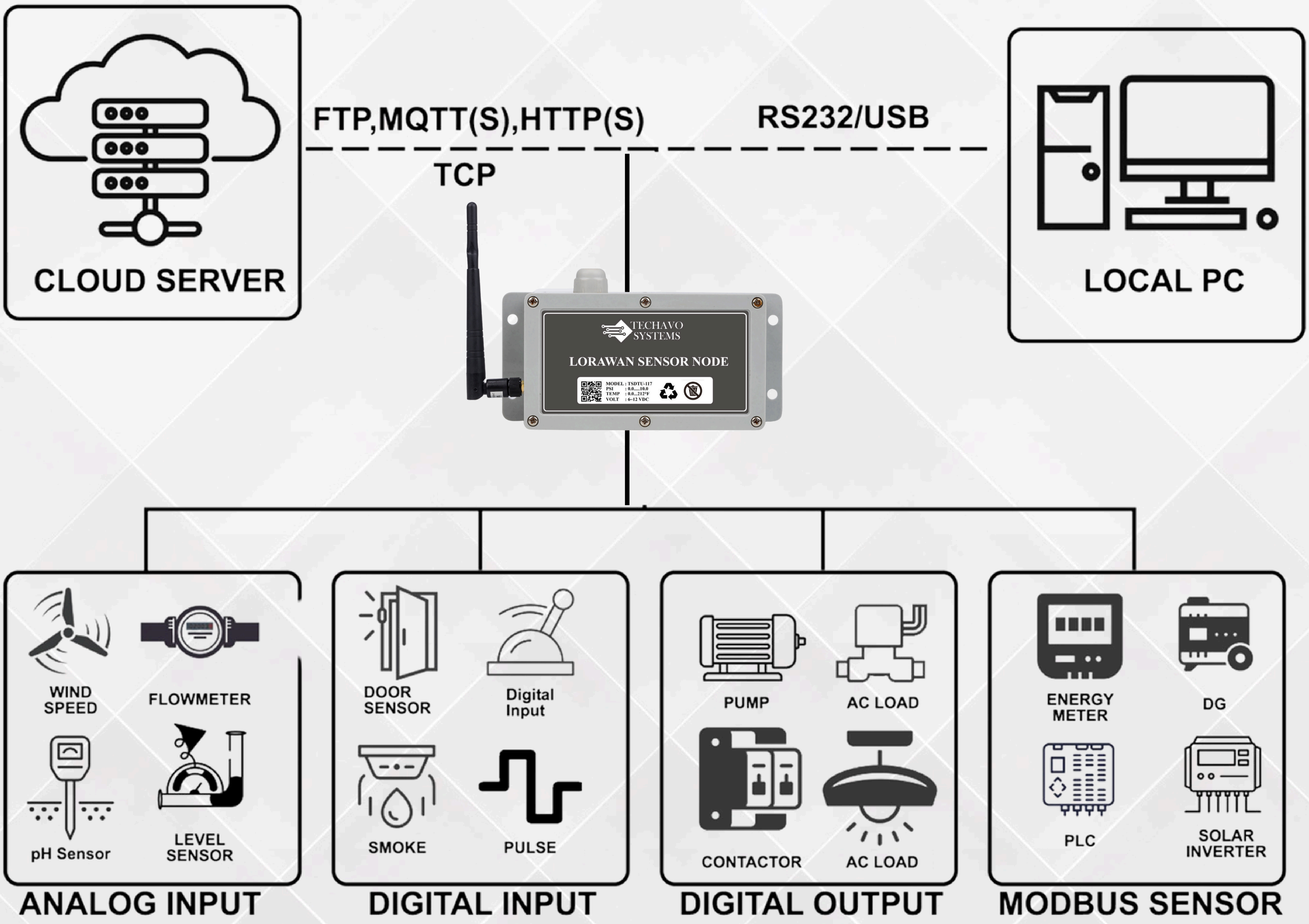
- STOP Mode: 2.7uA @ 3.3v
- LoRa Transmit Mode: 125mA @ 20dBm 44mA @ 14dBm

LoRa Side

- LoRa Chip: sx1276/sx1278
- 68 dB maximum link budget.
- +20 dBm - 100 mW constant RF output vs.
- +14 dBm high-efficiency PA.
- Programmable bit rate up to 300 kbps.
- High sensitivity: down to -148 dBm.
- Bullet-proof front end: IIP3-12.5 dBm.
- 127 dB Dynamic Range RSSI.
- LoRaWAN 1.0.3 Specification

Battery

- Li/SOCl2 unchargable battery
- Capacity: 4000mAh or 8500mAh
- Self Discharge: <1% / Year @ 25°C
- Max continuously current: 130mA
  Max boost current: 2A, 1 second



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APPLICATION AREA

- Wireless Alarm and Security Systems
- Home and Building Automation
- Automated Meter Reading
- Smart City
- Long-range Irrigation Systems
- Industrial Monitoring and Control

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